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Final Report

Summer 2026

ICT 361 001- Introduction to Robotics

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I. Introduction

This project will demonstrate how to build a robot that can follow a line on the floor automatically but also “see” the obstacle in front of it and make decisions based on that. It does this using a camera, some image processing, and a couple of extra sensors. The physical robot itself was already built beforehand; therefore, this project focuses on writing the code that controls it: the line detection, obstacle recognition, decision-making, and communication with the motors.

II. System Architecture

1. Hardware

- Raspberry Pi Camera
- Arduino/ESP32
- Ultrasonic Distance Sensor
- DC Motors
- Raspberry Pi

2. Software

- C++
- Python
- Flask
- OpenCV
- Numpy
- ONNX Runtime
- PySerial

III. System Implementation

1. Camera Initialization

The camera runs at 640×480 , which is enough resolution for reliable line and color detection without slowing the pipeline down.

2. Serial Communication

The Raspberry Pi and the Arduino talk to each other over USB serial at 115200 baud. The Raspberry Pi sends movement commands such as Forward, Left, Right, Stop and the Arduino continuously transmits ultrasonic sensor readings back to the Raspberry.

3. Line Detection

The robot only looks at the bottom part of the camera frame to find the line, since that's the part closest to the wheels. That part of the image is converted from BGR to HSV, which makes it easier to isolate a specific color. A mask is created to pick out just the black line and ignore everything else. From that mask, the code finds the center point of the line in that frame.

4. Line Searching

When the robot losing the line, it stops briefly and starts rotating left to find back the line. If that doesn't turn anything up within a set amount of time, it switches direction and tries rotating right instead, alternating back and forth until the line shows up again.

5. Obstacle Detection

The ultrasonic sensor is polled continuously, and the moment a reading comes back under 35 cm, the robot stops for 3 seconds and shifts into obstacle-analysis mode.

6. Color Detection

Once it stopped, the robot converts the full frame to HSV and checks it against color ranges for red and green. Red gets two separate ranges since it sits at both ends of the hue spectrum in HSV, which is a common quirk to work around. Whichever color has more matching pixels above the threshold wins; if neither clears the threshold, the obstacle is logged as other obstacle.

7. Obstacle Avoidance

- **Red obstacle:** the robot swings around it from the right (turn right, forward, turn left, forward, turn left, forward) and go back to the line.
- **Green obstacle:** the robot swings around it from the left (turn left, forward, turn right, forward, turn right, forward) and go back to the line.
- **Other obstacle:** robot stop.